

Selection of the DER $Q(V)$ dead-zone half-width

2026-08-02

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1 The parameter and the question

The distribution-connected DER follow a local reactive-power droop $Q(V)$ with a dead zone of half-width δ about a re-anchored voltage reference: the droop is inactive while $|V - V_{\text{anchor}}| < \delta$ and engages beyond it. δ is therefore an *amplitude threshold on the controller's input*, and it is a property of the characteristic rather than of any operating point.

Two effects are asserted to bound it. Too narrow, and the droop answers every small variation: the DER move continuously and the outer online feedback optimisation (OFO) repeatedly re-anchors against them. Too wide, and the droop is inactive between dispatches, so no local support acts and the interface reactive power drifts until the next OFO step.

This chapter reports three experiments. The first — sweeping δ against annual profile windows — is the obvious design and **it fails**; why it fails is itself a result (§4). The second measures the mechanism the dead zone governs rather than its consequence (§5). The third measures the threshold directly by controlled excitation (§6) and yields the design rule of §7.

1.1 Plant, actuators, controlled outputs

- **Plant.** Modified IEEE 39-bus transmission system, scenario `rural_700`, with four 110 kV distribution underlays. DSO₃ carries $2\times$ installed DER and $2\times$ active-load base; its reactive load is not scaled. Every DSO has zero constant reactive load and follows $Q_{\text{load}}(t) = 500 \text{ Mvar} \cdot m_{\text{qload}}(t)$.
- **Actuators.** DER reactive power (continuous, $Q(V)$ droop with dead zone), coupler three-winding OLTCs, machine two-winding OLTCs, MSC/MSR switched shunts.
- **Controlled outputs.** Reactive power at the EHV–HV interfaces, and nodal voltages within their band.
- **DER capability.** The physical VDE-AR-N-4120 diagrams throughout. Reactive capability is contingent on active power, which matters for window selection: a park at $P = 0$ has none (§3.1).
- **Controllers see only cached sensitivities** and their own measurements. No controller is given the plant equations.

1.2 Excitation

Two disturbance classes are used, and **no contingency is applied anywhere**: every run records `contingencies = []`, `measurement_noise.enabled = False` and `enable_reachability_guard = False`.

Profiled operation. The annual load/DER time series. This is the small-signal regime the dead zone is meant to filter.

Exogenous load step. A step multiplying every active-load profile from a fixed instant. Applied to the *interpolated* profile frame, so it is a true step at the 20 s dispatch resolution — perturbing the 15 min source would have been smeared into a ramp by interpolation. It reaches both plants through paths they already support and is *not* a contingency: no element is switched.

The absence of measurement noise bounds every conclusion below. A dead zone’s textbook justification is rejection of noise on the measured voltage; that mechanism is absent by construction, so the narrow-side penalty reported here is carried by genuine profile variation alone. Adding noise should penalise narrow dead bands further, making every δ^* reported here a *lower* bound on the value appropriate to a noisy plant. The direction is argued, not measured.

2 Provenance: the sensitivity reduction was corrected

Each controller acts on cached sensitivities derived from a Ward-reduced network. On 2026-08-01 that reduction was found to be wrong in three independent ways, and every result in this chapter postdates the correction.

Verified with `tools/check_reduction_fidelity.py`, which asks whether each reduced network, solved alone, reproduces the combined solution at the buses it retains — the property the

Defect	Effect
DSO reductions solved on the wrong power-flow branch	0.10 p.u. to 0.36 p.u. from the combined solution; t
Sub-network’s own auxiliary buses dropped	132 MW of the DSO’s own injection missing
Non-slack boundary couplers pinned at $P = 0$	multi-coupler DSOs forced their whole real exchan
TSO zones used generator <i>setpoints</i>	180 MW phantom injection under distributed slack

Table 1: Defects in the reduced-network sensitivities, corrected 2026-08-01. All four inputs are locally measurable: PCC flows, own topology, own machine telemetry.

reduction claims. Convergence is *not* that test: a reduced net with stranded boundary flows converges happily to a different operating point.

After correction all sixteen DSO reductions ($4 \text{ DSOs} \times 4 \text{ windows}$) reproduce the combined solution to **0.000000** p.u.; the TSO zones improved up to $5\times$ and retain a residual of 0.012 p.u. to 0.018 p.u., which is a linearisation-point offset in a smooth regime and is accepted.

The correction is not cosmetic: at 2016-01-05 08:00, $\delta = 0.005$, the interface- Q metric moved 0.4410 Mvar $\rightarrow$ 0.4929 Mvar ($\approx 12\%$). Runs record `sensitivity_reduction_rev`, and the admission filter requires rev 2, so pre-correction runs cannot enter these curves.

3 Windows

Five live windows spanning net infeed -117 to $+2200$ MW, plus one retained as a null result.

Window	DER P [MW]	Load P [MW]	Net infeed [MW]	Parks with zero Q cap.
2016-02-22 13:00	2262.4	2379.9	-117.4	16/44
2016-01-05 08:00	2324.5	1916.0	408.6	16/44
2016-01-15 03:00	2605.7	1800.7	805.0	16/44
2016-12-18 14:00	2741.8	1374.7	1367.1	16/44
2016-05-01 16:00	3458.6	1258.6	2200.0	–
2016-07-15 03:00	29.2	1055.3	-1026.1	44/44

Table 2: Windows characterised by current-topology quantities. The last is degenerate (§3.1).

Three points of method:

The screening figures that selected the windows are void. Three windows were drawn from a Tier-1 seasonal screening measured on an *earlier topology*. That ranking does not transfer: the window it called the annual maximum turned out to be the least demanding of the set. The figures are recorded as provenance only and are never quoted as properties of these runs.

A deep-import window with capable DER does not exist in this data. DER output and load are positively correlated, so the hours with reactive capability are the high-generation hours. Screening all 35 136 quarter-hours, *no* hour combines $P_{\text{DER}} \geq 2200$ MW with net infeed below -300 MW. The import side of the axis therefore stops at -117 MW, and any statement about the voltage-drop regime rests on that single window.

Feasibility is empirical. 2016-05-01 16:00 aborted with a non-convergent power flow under rev 1 and converges under rev 2, because removing the phantom generation of §2 gave its reduced net a solution. Offline screening did not predict this in either direction.

3.1 A degenerate window

At 2016-07-15 03:00 — July, 03:00, no photovoltaic infeed and a multi-day wind lull — aggregate DER infeed is 29.2 MW. Under the VDE diagram *all* 44 parks report zero reactive capability and both zone PCC capability intervals collapse to $[0, 0]$ Mvar. With no DER able to move, the $Q(V)$ characteristic never binds: two dead bands an order of magnitude apart return identical results to six digits.

This is a property of the operating point, not a defect, and it bounds the design rule: **dead-band selection presupposes DER reactive headroom**. The window is excluded from every optimum and retained as a null result. It also serves as a determinism control — because δ provably cannot act there, its identical outputs prove the pipeline is deterministic, which matters for §4.

4 Experiment 1: the profiled sweep, and why it fails

Full factorial, $\delta \in \{0, 0.001, 0.0025, 0.005, 0.0075, 0.01, 0.015, 0.02, 0.03\}$ p.u. crossed with the five live windows: 45 closed-loop co-simulations against the PowerFactory RMS plant, each with a quasi-static reference leg.

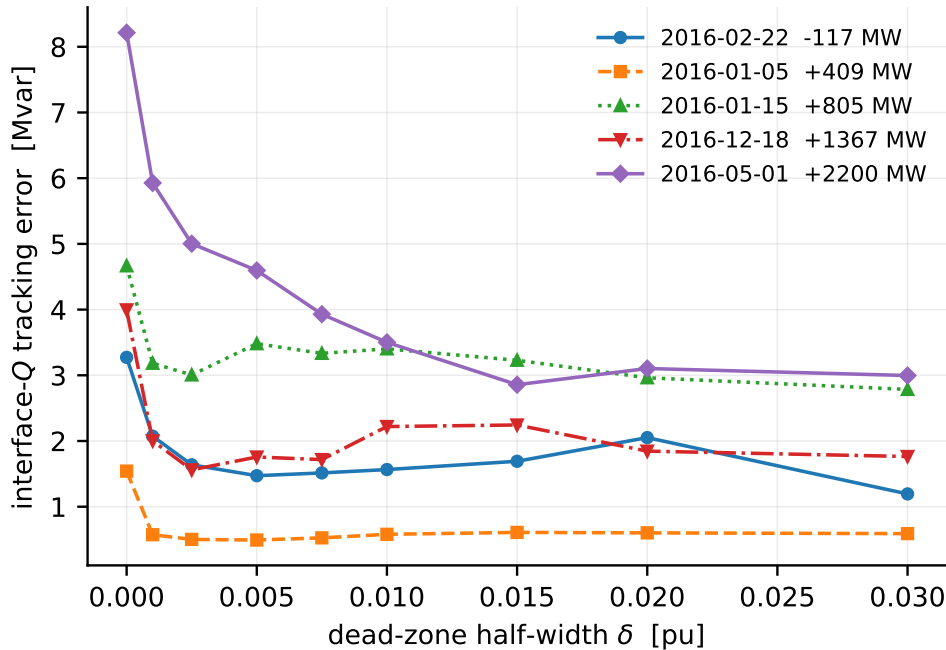


Figure 1: Interface- Q tracking error against dead-zone half-width, one curve per window. Only the mildest window (+409 MW) gives a clean interior minimum; the stressed windows are non-monotone.

4.1 Result: no optimum can be defined

- No δ is **Pareto-optimal in every window** over (interface Q , DS voltage) — the intersection is empty (Fig. 3).
- The interface- Q argmin scatters over $0.0025 \dots 0.03$ with no relation to loading: $CV = 0.715$ across the five windows.
- The stressed windows' curves are non-monotone with 20% to 25% scatter.

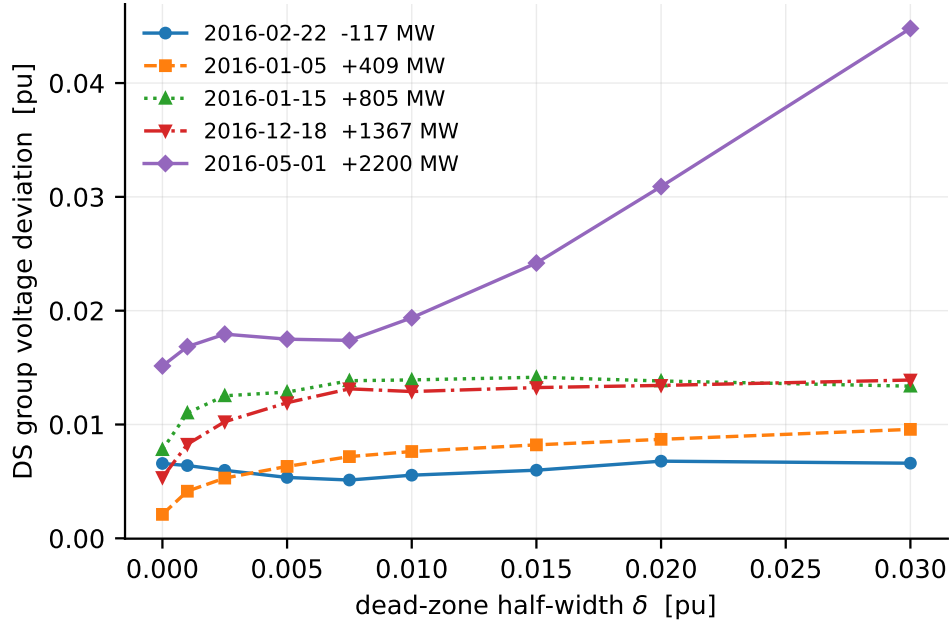


Figure 2: DS group voltage deviation against δ . Every exporting window is best at $\delta = 0$; the importing window (-117 MW) has an interior optimum, i.e. the trade-off reverses sign with the direction of interface flow.

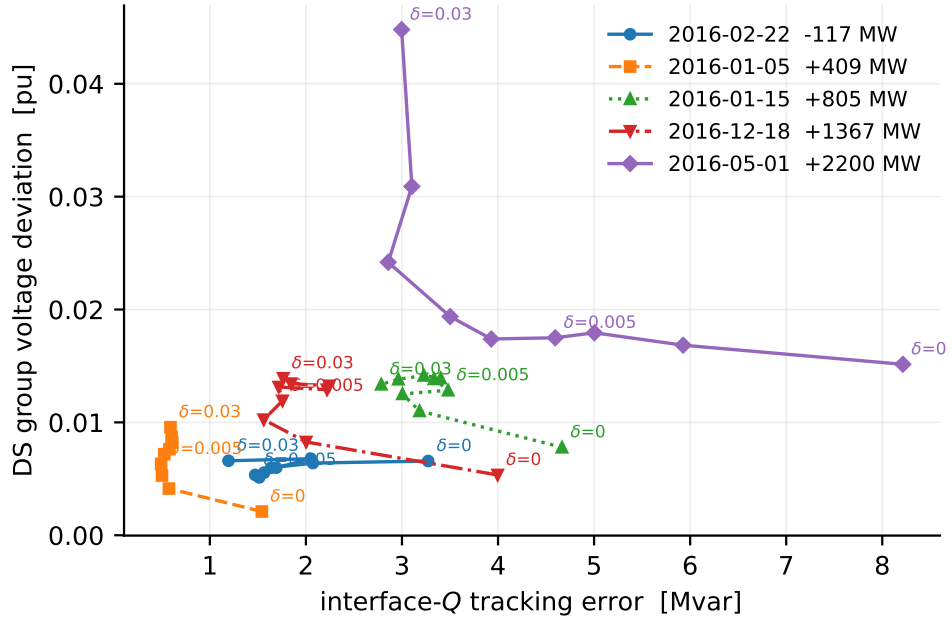


Figure 3: The two controlled quantities against each other, with δ as the path parameter. A monotone trade-off would trace a simple curve; the stressed windows fold back on themselves.

4.2 Why: the metric is not a function of δ

The scatter is *structural*, not statistical. The degenerate window (§3.1) reproduces to six digits, so the pipeline is deterministic; different δ therefore reach genuinely different equilibria. This is the dead-zone multi-equilibrium behaviour — the same non-uniqueness identified between the quasi-static and RMS representations — here dominating the stressed operating points rather than appearing as isolated anomalies.

Four design flaws compound it: one realisation per cell, so a single trajectory samples one basin arbitrarily; a δ -dependent initial condition, because the $Q(V)$ seeding uses the dead band; a metric mixing transient with steady state; and no measurement of the mechanism the dead band actually governs.

Taking an argmin over a bistable metric is not a valid selection procedure, and the remainder of this chapter does not attempt it.

Two findings from this experiment do survive, because they do not depend on locating a minimum:

1. **δ must be non-zero.** At $\delta = 0$ the interface- Q error is roughly $3\times$ its best value in every window. The collapse is abrupt: $\delta = 0.001$ p.u. already recovers about 90 % of it.
2. **The voltage trade-off reverses with the direction of interface flow** (Fig. 2). Exporting windows want $\delta = 0$ for DS voltage; the importing window has an interior optimum at 0.0075 p.u..

5 Experiment 2: measuring the mechanism

The dead zone governs how much the local droop *moves*. Counting actuator motion rather than reading a possibly-bistable end state gives a metric that cannot be confounded by which equilibrium a run settles into.

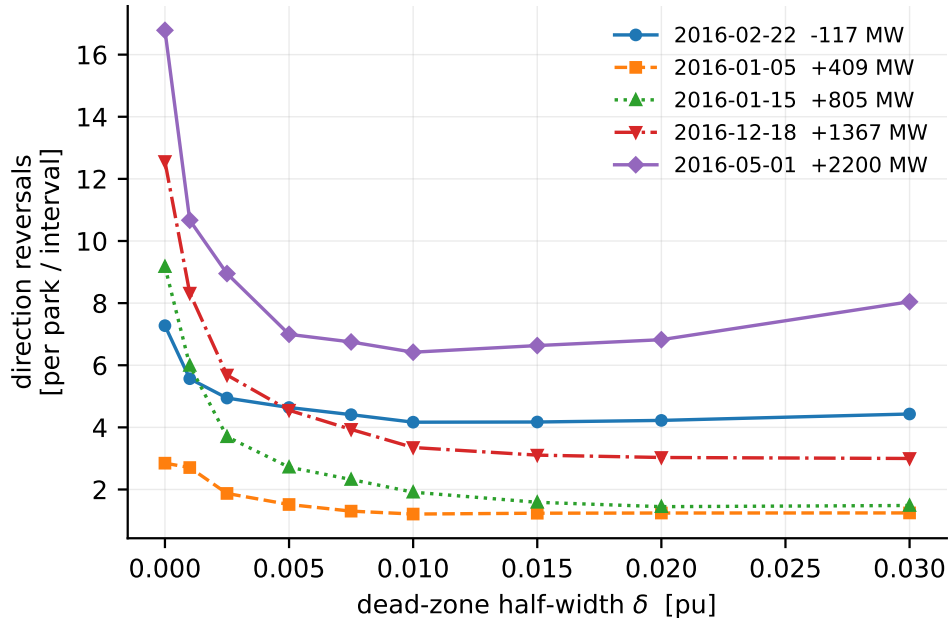


Figure 4: Direction reversals of the DER reactive power, per park per dispatch interval, against δ . Monotone with a clear knee in every window — compare the same runs in Fig. 1.

Defining the knee as the smallest δ within 10 % of that window’s floor:

	mean	CV
chatter knee	0.0100 p.u.	0.418
interface- Q argmin	0.016 p.u.	0.715

The chatter knee is about twice as well determined, and it measures the quantity δ directly controls. It is still a $3\times$ range rather than a number, which motivates §6.

A normalisation hypothesis, tested and refuted. If δ were simply a threshold relative to local voltage variability, normalising by the measured σ_V (which ranges 0.00128 p.u. to 0.00481 p.u. across the windows) should collapse the curves. It does not: the knee CV goes $0.418 \rightarrow 0.427$ and the argmin CV $0.715 \rightarrow 0.747$. Both marginally *worse*. δ is not a constant multiple of σ_V .

6 Experiment 3: the threshold, measured directly

Rather than infer the threshold from 300s of uncontrolled multi-frequency excitation, apply one disturbance of known amplitude and find the δ at which the droop stops responding. One operating point (2016-01-05 08:00 — the only window whose response is unambiguous, and the lowest σ_V , so the step dominates the background).

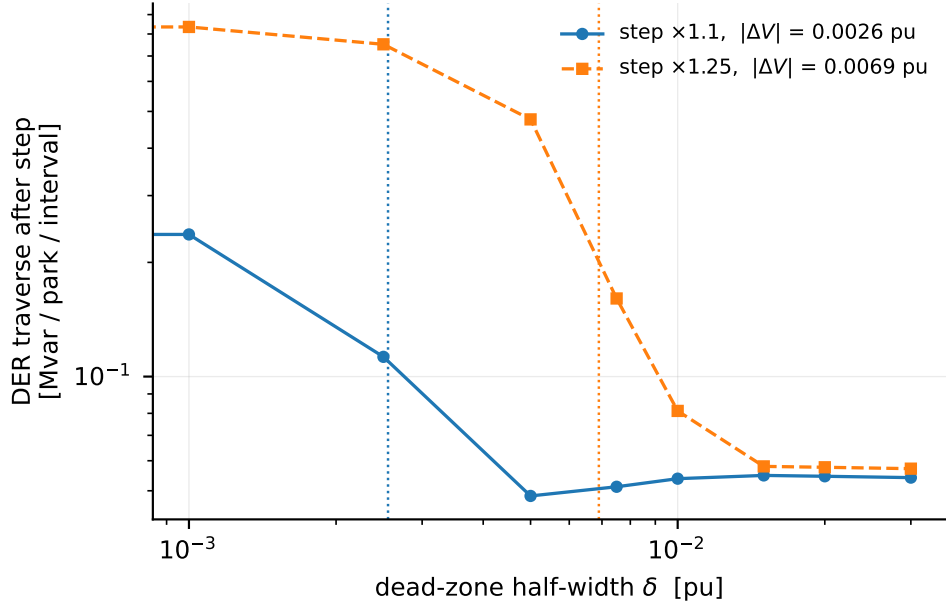


Figure 5: Post-step DER traverse against δ at two excitation amplitudes. Dotted verticals mark each amplitude’s open-loop $|\Delta V|$. The response collapses to the quiet floor as δ crosses the excitation.

δ [p.u.]	0.0025	0.005	0.0075	0.01	0.015	0.02
$\times 1.10$, $ \Delta V = 0.00255$ p.u.	0.113	0.049	0.051	0.054	0.055	0.055
$\times 1.25$, $ \Delta V = 0.00690$ p.u.	0.752	0.476	0.161	0.081	0.058	0.058

Table 3: Post-step DER traverse [Mvar per park per interval]. Bold marks the first δ at which the response has essentially reached the quiet floor. In both rows that happens as δ passes the excitation amplitude.

At $|\Delta V| = 0.00255$ p.u. the response falls to the floor between $\delta = 0.0025$ and 0.005 ; at $|\Delta V| = 0.00690$ p.u. between $\delta = 0.005$ and 0.0075 . **In both cases the transition brackets the excitation amplitude**, so the engagement threshold satisfies

$$\delta_{\text{engage}} \approx |\Delta V|, \quad k = \delta_{\text{engage}}/|\Delta V| \approx 1, \quad (1)$$

at two independent amplitudes differing by a factor of 2.7. The dead zone behaves as specified: the droop acts when the deviation exceeds δ and is silent when it does not.

The excitation is not independent of δ . Measured $|\Delta V|$ at $\times 1.25$ runs from 0.00153 p.u. at $\delta = 0$ to 0.00690 p.u. at $\delta \geq 0.015$: a tight dead band makes the droop act, and acting suppresses the very deviation it responds to. The open-loop excitation must therefore be read at wide δ , where the droop is inactive. *The dead band partly determines its own input*, which is a closed-loop property worth stating explicitly.

The threshold is measured on the TS parks. Equation (1) was obtained from the aggregate traverse, which the TS-connected parks dominate by a factor of 10 to 40; it is therefore in effect a TS result, and it does not by itself license one δ for both voltage levels. Resolving the excitation per park at wide δ (open loop, $\times 1.25$) gives

	n	median $ \Delta V $ [p.u.]	max [p.u.]	max/median
TS parks	4	0.00645	0.00813	1.26
DS parks	40	0.00673	0.00907	1.35

The two populations differ in median by under 4%, and neither spreads by more than $1.35\times$ about its own median. Referenced to the *maximum* park deviation rather than the mean — the appropriate reference, since the dead band must be chosen against the worst park it has to keep silent — $k \approx 1$ holds for both levels. A single uniform δ is therefore justified *for this disturbance class*; whether it survives a disturbance that loads one level preferentially is a separate question, taken up in §7.

Large dead bands are unreachable by *uniform* disturbances. A static scan gives the load step needed to induce a given deviation: under a uniform increase at all profiled buses, exceeding 0.02 p.u. requires roughly *tripling* system load ($4938 \rightarrow 8770$ MW). That is a stress test, not an event.

This conclusion is conditional on the disturbance being uniform, and that qualification matters: a load step *concentrated* at a single weak bus is far more efficient. At the most severe distribution bus, 43 MW suffices for 0.02 p.u. — some 90 times less energy than the uniform step, with a spatial spread of 0.046 p.u. across the parks. A localised event of entirely ordinary size therefore reaches deviations that the uniform scan places beyond reach. The claim that should be carried forward is the narrower one: *under uniform loading* at this operating point, $\delta \gtrsim 0.01$ leaves the local droop effectively disabled. Amplitudes at and beyond $\times 1.5$ were attempted and proved unusable for a different reason: they destabilise the cascade (traverse 80.7 against a floor of 0.057 , and a settling-gate failure), so they measure instability rather than threshold.

7 The design rule

Equation (1) converts δ from a tuning parameter into a stated design choice:

Choose δ equal to the voltage deviation the local layer should ignore. The droop then engages for anything larger and is silent for anything smaller.

The performance results then bound the admissible range rather than selecting within it:

Bound	From	Value
lower	$\delta = 0$ costs $3\times$ in interface- Q tracking	$\delta > 0$
lower	chatter knee (§5)	$\delta \gtrsim 0.005$
upper	DS voltage degrades monotonically (exporting)	as small as tolerable
upper	droop never engages on realistic events	$\delta \lesssim 0.01$

7.1 Worked example: a 0.01 p.u. limit

Suppose the requirement is that the local layer must act before the deviation reaches 0.01 p.u.. By (1), engagement occurs when the deviation exceeds δ , so the requirement is

$$\delta \leq 0.01 \text{ p.u..}$$

$\delta = 0.01$ p.u. is the largest admissible value and satisfies the requirement exactly at the limit. It also sits at the chatter knee, so it captures essentially all of the achievable chatter reduction.

It is, however, the *worst* admissible choice on the other two measures. At the reference window, relative to their best values, $\delta = 0.01$ p.u. costs 18 % in interface- Q tracking (0.580 against 0.493 Mvar) and 73 % in step rejection (peak excess 0.335 against 0.194 Mvar), and it gives up a factor of 3.6 in DS voltage against $\delta = 0$.

Recommendation: $\delta = 0.005$ p.u.. It satisfies the 0.01 p.u. requirement with a factor-of-two margin, and it is where three independent criteria at the reference window already coincide:

Criterion	δ
profiled interface- Q optimum	0.005
step-rejection optimum	0.005
chatter knee	0.0075

8 The dead band as a detector threshold: N-1 evidence

Provisional. Every number in this section comes from ONE operating window (2016-01-05 08:00, net infeed 409 MW), 24 RMS co-simulations run 2026-08-02/03. The reasoning and the method are settled; the *values* await replication at further windows and will be restated then.

8.1 Why the earlier sections could not find an optimum

The Q(V) layer exists to reject *events*. Profile-driven voltage shifts are the OFO’s business: it re-optimises every 180 s (TSO) or 20 s (DSO) and re-anchors each park’s V_{ref} when it writes a setpoint. The droop compares $|V - V_{\text{anchor}}|$ against δ , so δ does not discriminate voltage *levels* — it discriminates *drift since that park’s last dispatch*.

That makes δ a **detector threshold**, and a detector has no optimum. It has an admissible interval, bounded below by the disturbance it must ignore and above by the one it must catch:

$$\underbrace{\max_{\text{profile}} |\Delta V|}_{\text{OFO's business}} \lesssim \delta \lesssim \underbrace{\min_{\text{event}} |\Delta V|}_{\text{droop's business}} . \quad (2)$$

This is why δ^* scattered with CV = 0.715 across windows in §4: the argmin was being fitted to a curve whose flat region *is* the answer.

8.2 Method

The disturbance is a synchronous-machine trip — a credible N-1 — delivered to the RMS plant as an `EvtOutage` on the machine and its unit transformer. Horizon 600 s; the trip fires at 200 s, i.e. 20 s *after* a TSO dispatch, which maximises the droop-only exposure to 160 s. Each of the eight dead bands is run three times: undisturbed twin, mild trip, severe trip. Every deviation is referenced to the same- δ twin; the pre-trip deviation between a run and its twin is **identically zero**, so the referencing contributes no artefact.

The disturbance is a transient, and that is the point. The excursion is not a sustained offset. At the worst park, with the droop disabled, a gen-7 trip gives

t [s]	179	190	195	220	260	300	600
V [p.u.]	1.0441	0.9132	0.8949	0.9024	1.0052	1.0389	1.0559
$ \Delta V $ [p.u.]	0.0000	0.1315	0.1497	0.1424	0.0397	0.0060	0.0093

(the twin holds 1.0441 p.u. to 1.0466 p.u. throughout, so the deviation is attributable to the trip alone; pre-trip $|\Delta V|$ is identically zero). The peak of 0.2240 p.u. occurs 10.5 s after the trip; $|\Delta V|$ exceeds 0.05 p.u. for 90 s and 0.01 p.u. for 118 s, after which the control stack has restored the voltage and the *settled* deviation is below 0.01 p.u..

That timescale is exactly why the dead band matters. The TSO re-optimises every 180 s; the whole excursion opens and closes *inside one TSO interval*. Whatever is done about it in that window is done by the DSO layer and by the local droop — which is the inter-OFO-step role the dead band governs.

Quasi-static screening cannot select the severe case. A static outage scan ranks the two machines in the opposite order to the dynamic plant:

worst-park $ \Delta V $ [p.u.]	static scan (open loop, settled)	RMS, droop disabled (closed loop, peak)
trip gen 1 (650 MW)	0.0854	0.1039
trip gen 7 (830 MW)	0.0122	0.2240

The two columns are different quantities and must not be read as a ratio: the static scan is an unregulated power flow after the outage, while the RMS column is the peak of a closed-loop transient. The comparison that matters is the *ordering*. Statically the larger machine looks like the milder disturbance — a distributed slack absorbs its 830 MW instantaneously and the post-outage power flow barely moves. Dynamically it produces the deeper swing and is the only case reaching an undervoltage violation. A severe case selected on the quasi-static plant would therefore have been the wrong machine.

This is a concrete instance of the general point of §2: the controllers may reason on a reduced or quasi-static model, but the *evidence* must come from the plant that has the dynamics.

8.3 The lower bound: what the droop must ignore

Profile-driven drift is measured on the twins whose dead band is provably never crossed ($\delta \geq 0.025$, four runs, $n = 48$ TS and 3360 DS inter-dispatch windows). Admission is judged against the *widest* twin’s maximum, never against a twin’s own: a droop that successfully suppresses drift below its own threshold would otherwise certify itself as inactive.

	p90 [p.u.]	max [p.u.]	windows exceeding δ
TS parks	0.00285	0.00304	—
DS parks	0.00108	0.01101	—
$\delta = 0.001$	28/48 TS, 499/3360 DS		
$\delta = 0.005$	0/48 TS, 20/3360 DS (0.6 %)		
$\delta = 0.01$	0/48 TS, 4/3360 DS (0.12 %)		
$\delta = 0.025$	0/48 TS, 0/3360 DS		

Table 4: Profile-driven drift and the resulting false-activation rate. The DS distribution carries a $10\times$ tail (p90 0.0011, max 0.0110): the DS bound is set by rare windows, not by typical operation.

δ [p.u.]	compensation gen 1	$1 - \hat{V}/\hat{V}_{\text{no droop}}$ gen 7	min. absolute V [p.u.] gen 1	gen 7
0	0.78	0.55	1.006	0.935
0.005	0.73	0.54	1.003	0.934
0.01	0.67	0.54	0.996	0.933
0.025	0.49	0.57	0.972	0.939
0.05	0.28	0.52	0.966	0.916
0.075	0.08	0.42	0.951	0.880
0.15	0.00	0.00	0.937	0.816

Table 5: Event rejection at the TS parks. With the droop disabled a gen-7 trip drives a park terminal to 0.816 p.u.; the Q(V) layer prevents that violation, but only while $\delta \lesssim 0.05$.

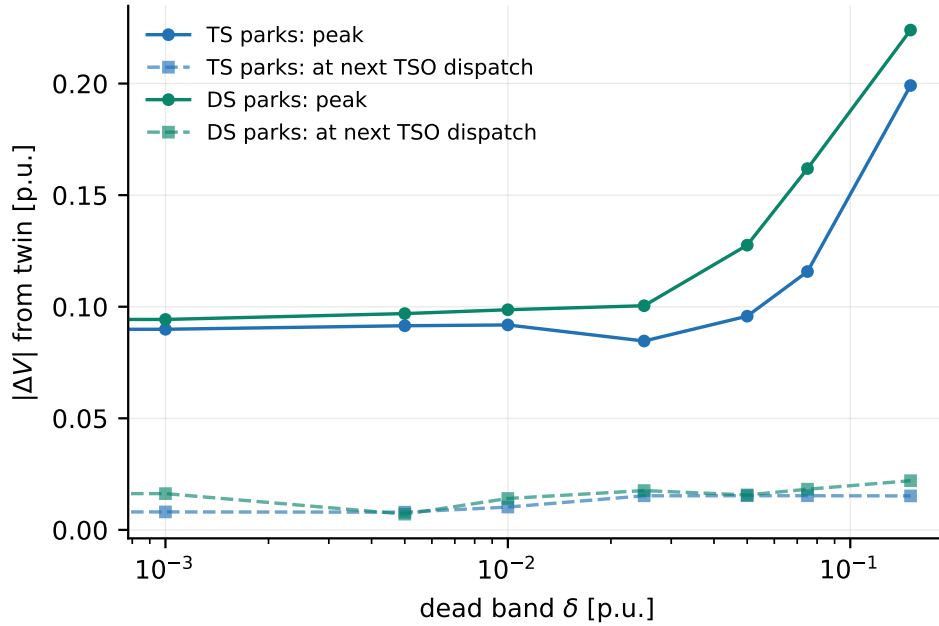


Figure 6: Post-trip peak deviation and the residual still standing when the next TSO dispatch acts, against δ (gen 7). The knee lies above 0.01 p.u..

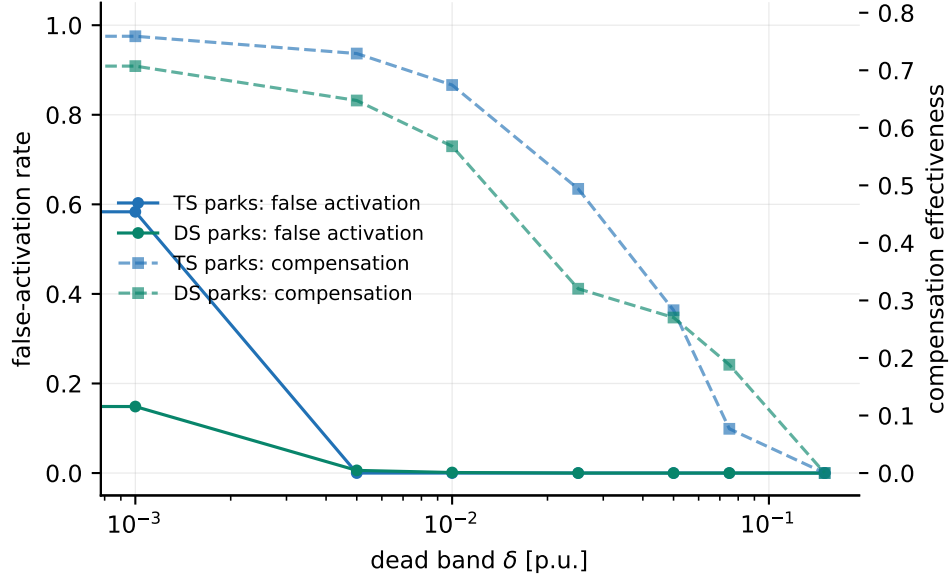


Figure 7: The two competing rates against δ : false activation on profile drift (left axis) and compensation effectiveness on the event (right axis). An admissible δ lies where the first has fallen to zero and the second has not yet decayed.

8.4 The upper bound: what the droop must catch

8.5 Choice of δ

Both bounds of (2) are now measured, and they do not conflict: the admissible interval is non-empty. Within it, δ is a quantile choice.

The recommendation of §7, $\delta = 0.005$ p.u., is confirmed by this independent experiment, on three grounds:

1. **It precedes the knee.** Peak deviation is flat below 0.005 p.u. and rises steeply above 0.01 p.u. — +20 % from $\delta = 0.005$ to 0.01, then +55 % to 0.025 (Fig. 6).
2. **It retains almost all achievable rejection:** compensation 0.73 (gen 1) and 0.54 (gen 7) against ceilings of 0.78 and 0.55 at $\delta = 0$ — 94 % and 98 % of what any dead band can deliver — while $\delta = 0.025$ retains only 0.49 on gen 1.
3. **It is effectively silent on profiles:** no TS window and 0.6 % of DS windows activate the droop.

Why a quantile and not the maximum. Zero false activation on the DS parks requires $\delta \geq 0.025$ p.u., because the DS drift *maximum* is 0.0110 p.u.. That bound should not be used: a maximum grows with the observation horizon — the same window sampled over 300 s yields a DS maximum of 0.0014 p.u., against 0.0110 p.u. over 600 s — so a rule anchored to it is a rule anchored to how long one watched. A quantile is stable under that choice. Stated in those terms:

At $\delta = 0.005$ p.u. the local layer ignores 99.4 % of profile-driven inter-dispatch variation, retains 94 % to 98 % of the event compensation achievable at any dead band, and holds the worst credible N-1 at 0.934 p.u..

One dead band, not two. The TS admissible set ($[0.005, 0.075]$) contains the DS set ($[0.025, 0.075]$) at this operating point: the DS parks bind and the TS parks have margin.

Separate δ_{TS} and δ_{DS} are therefore not required by this evidence. The per-level machinery exists and a $\delta_{\text{TS}} \times \delta_{\text{DS}}$ sweep is the natural follow-up, but the null result stands until it is run.

9 Threats to validity

1. **No measurement noise** (§1.2). The mechanism a dead zone classically exists to reject is absent, so every δ^* here is expected to be a lower bound for a noisy plant. This is the single change most likely to move the answer.
2. **Equation (1) is verified at $|\Delta V| = 0.0026$ and **0.0069** p.u..** Applying it at 0.01 p.u. extrapolates by about $1.5\times$. The extrapolation is modest and the relation is linear by construction, but it is unverified at that amplitude, and amplitudes large enough to test it directly destabilise the cascade.
3. **One operating point for the threshold.** Whether $k \approx 1$ holds across loading is untested; the profiled windows validate the *consequences* of a choice, not the threshold itself. Repeating §6 at a second window would settle it.
4. **Purposive windows, $n = 5$.** No frequency-weighted or annual-average statement is supportable. The set is also asymmetric about zero for the structural reason of §3.
5. **Contingencies: addressed, at one window (§8).** A trip changes the topology, and therefore the controllers' cached sensitivities, whereas a load step changes only the operating point. Sections 4–7 rest on load steps alone; §8 adds N-1 evidence and reaches the same $\delta = 0.005$ p.u. by an independent route. That evidence is **provisional**: one operating window, two machines.
6. **The N-1 lower bound rests on a tail.** The DS false-activation rate at $\delta = 0.005$ p.u. is 0.6% — 20 windows out of 3360. A bound set by twenty samples is sensitive to which windows are drawn, which is the strongest single argument for replicating §8 across operating points before quoting the interval as general.
7. **Two machines, not a contingency list.** §8 trips two of six generators; **gen 9** is excluded because the static screening diverges on it. No N-1 *completeness* claim is made, and line outages — which the RMS adapter does not yet deliver — are untested.
8. **Residual TSO reduction error of 0.012 p.u. to 0.018 p.u. (§2).** The sensitivities are linearised once, and measurements come from the combined model, so an offset in a smooth regime is tolerable — unlike the pre-correction DSO error, which placed the linearisation on the wrong branch of the P - V curve.
9. **The multi-equilibrium behaviour is characterised only qualitatively.** An ensemble over perturbed initial conditions per cell would map the basins directly and is the natural next experiment.